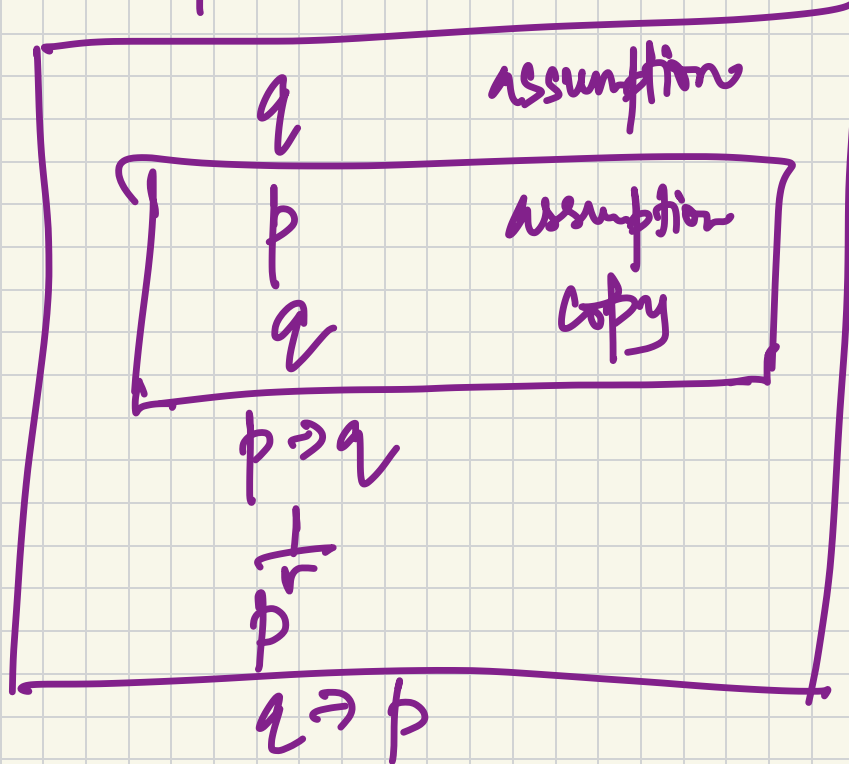


COL 703/7203

10 Aug

$\neg(p \rightarrow q) \vdash q \rightarrow p$

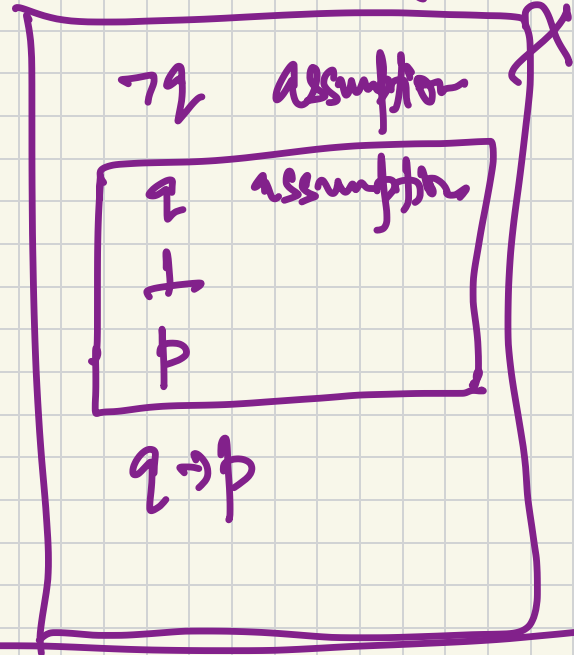
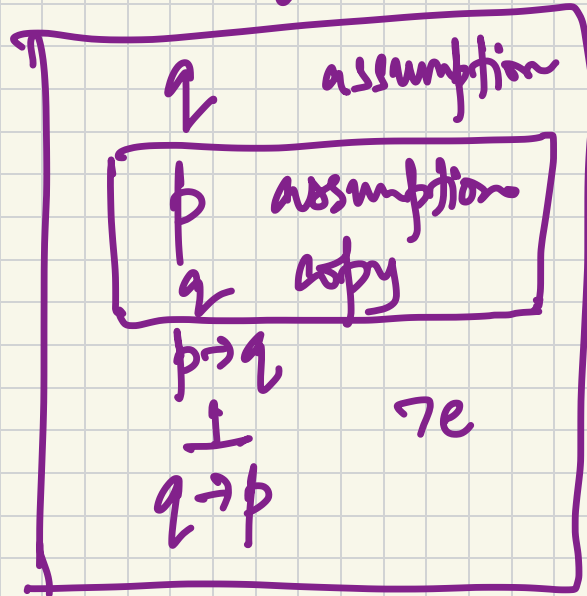


$$\neg(p \rightarrow q) \vdash q \rightarrow p$$

$$\phi \vee \psi \quad \boxed{\neg\phi} \quad \boxed{\neg\psi}$$

$$q \vee \neg q$$

LEM



$$q \rightarrow p$$

$q \vee \neg q$

q

$\neg \vee \neg \neg$

$\neg$
 ~~$\neg$~~

q

~~$\neg \neg$~~

$\neg q$

$\neg \vee \neg \neg$

$\neg q$

~~$\neg$~~

$\neg q$

~~$\neg \neg$~~

Extra
information
($\neg p$ or $\neg \neg p$)
should not
hurt the
proof.

Semantic Equivalence

$$\phi \equiv \psi$$

$$\phi \models \psi$$

$$\psi \models \phi$$

The brute force way to prove this is through truth tables.

One can, of course, appeal to soundness and completeness of prop. logic, and prove the validity of $\phi \models \psi$ and $\psi \models \phi$.

But we can also manipulate the expressions through known equivalences.

$$\begin{aligned}\phi \vee \phi &\equiv \phi \\ \phi \wedge \phi &\equiv \phi\end{aligned} \quad \left| \text{Idempotence} \right.$$

$$\begin{aligned}\phi \wedge \psi &\equiv \psi \wedge \phi \\ \phi \vee \psi &\equiv \psi \vee \phi\end{aligned} \quad \left. \vphantom{\begin{aligned}\phi \wedge \psi &\equiv \psi \wedge \phi \\ \phi \vee \psi &\equiv \psi \vee \phi\end{aligned}} \right\} \text{Commutativity}$$

$$\begin{aligned}(\phi_1 \wedge \phi_2) \wedge \phi_3 &\equiv \phi_1 \wedge (\phi_2 \wedge \phi_3) \\ (\phi_1 \vee \phi_2) \vee \phi_3 &\equiv \phi_1 \vee (\phi_2 \vee \phi_3)\end{aligned} \quad \left| \text{Associativity} \right.$$

$$\begin{aligned}\phi \wedge (\phi \vee \psi) &\equiv \phi \\ \phi \vee (\phi \wedge \psi) &\equiv \phi\end{aligned} \quad \left. \vphantom{\begin{aligned}\phi \wedge (\phi \vee \psi) &\equiv \phi \\ \phi \vee (\phi \wedge \psi) &\equiv \phi\end{aligned}} \right\} \text{Absorption}$$

$$\phi \wedge (\psi_1 \vee \psi_2) \equiv (\phi \wedge \psi_1) \vee (\phi \wedge \psi_2)$$

$$\phi \vee (\psi_1 \wedge \psi_2) \equiv (\phi \vee \psi_1) \wedge (\phi \vee \psi_2)$$

Distributivity

$$\neg \neg \phi \equiv \phi$$

(Double negation)

$$\neg (\phi \vee \psi) \equiv$$

$$\neg \phi \wedge \neg \psi$$

$$\neg (\phi \wedge \psi) \equiv$$

$$\neg \phi \vee \neg \psi$$

De Morgan's Laws

$$\phi \vee \neg \phi \equiv \text{true}$$

$$\phi \wedge \neg \phi \equiv \text{false}$$

$$\phi \vee \text{true} \equiv \text{true}$$

$$\phi \vee \text{false} \equiv \phi$$

$$\phi \wedge \text{true} \equiv \phi$$

$$\phi \wedge \text{false} \equiv \text{false}$$

$$\underbrace{(\phi \vee (\psi_1 \vee \psi_2))}_{\alpha} \wedge (\psi_2 \vee \neg \phi) \equiv \psi_2 \vee (\neg \phi \wedge \psi_1)$$

$$(\phi \vee \psi_1) \vee \psi_2$$

$$(\psi_2 \vee (\phi \vee \psi_1)) \wedge (\psi_2 \vee \neg \phi)$$

$$\alpha \quad \psi_2 \vee ((\phi \vee \psi_1) \wedge \neg \phi) \quad \begin{matrix} \beta & \gamma \\ \alpha \vee (\beta \wedge \gamma) \\ \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma) \end{matrix}$$

$$\psi_2 \vee (\neg \phi \wedge (\phi \vee \psi_1))$$

$$\psi_2 \vee ((\neg \phi \wedge \phi) \vee (\neg \phi \wedge \psi_1))$$

Normal Forms (of propositional logic formulas)

Negation Normal Form (NNF)

A ~~iff~~ is in negation normal form if it only uses $\wedge$, $\vee$, and literals (atoms or their negation).

Every ~~iff~~ is equivalent to one in NNF.

Convert $\neg(p \rightarrow (p \wedge q))$ into NNF.

$$\neg(p \rightarrow (p \wedge q))$$

$$\neg(\neg p \vee (p \wedge q))$$

$$\neg \neg p \wedge \neg(p \wedge q)$$

$$p \wedge (\neg p \vee \neg q)$$

$$(p \wedge \neg p) \vee (p \wedge \neg q)$$

Disjunctive Normal Form

conjunction

A fff is in disjunctive normal form if it is a disjunction conjunction of one or more terms where each term is a conjunction disjunction of one or more literals.

How do we derive these normal forms?
→ Manipulating the expression through
known equivalences

→ Truth table:

p	q	r	ϕ
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$\begin{aligned} & (\neg p \wedge \neg q \wedge \neg r) \\ \vee & (\neg p \wedge q \wedge r) \\ \vee & (p \wedge \neg q \wedge \neg r) \\ \vee & (p \wedge q \wedge r) \end{aligned}$$

CNF
exercise

$$\underbrace{(\neg q \vee p \vee r)}_{\text{true}} \wedge \underbrace{(\neg p \vee q \vee s)}_{\text{true}} \wedge \underbrace{(\neg r \vee \neg s \vee \neg q)}_{\text{true}}$$

↓

Claim A disjunction of literals
 $(l_1 \vee l_2 \vee \dots \vee l_m)$ is
 valid iff there are $1 \leq i, j \leq m$
 such that $l_i = \neg l_j$.

Proof.

Exercise